

Audio, Image, and Video Data

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GV330

Overview

1 Audio

2 Image

3 Video

Sound as Data

Sound waves (as data) are a combination of two things:

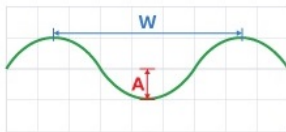
- Frequency: Pitch (low or high)
- Amplitude: Volume (low or high)



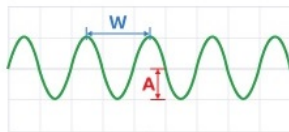
No sound



A loud sound with a low pitch



A quiet sound with a low pitch



A quiet sound with a high pitch



Amplitude



Wavelength

Audio Data Applications in Political Science

- What people say matters, but perhaps **how** they say it contains additional information
- **Audio recordings**: political debates, campaign advertisements, campaign speeches, legislative speeches, television news, talk radio, judicial deliberation.
- Challenges:
 - How to represent audio recordings numerically?
 - How to infer relevant quantities?
 - How to deal with temporal patterns (speech flow)?
- Method: **audio featurization**, requiring **preprocessing** into **utterances**
- See R packages `tuneR`, `seewave`, and `communication`

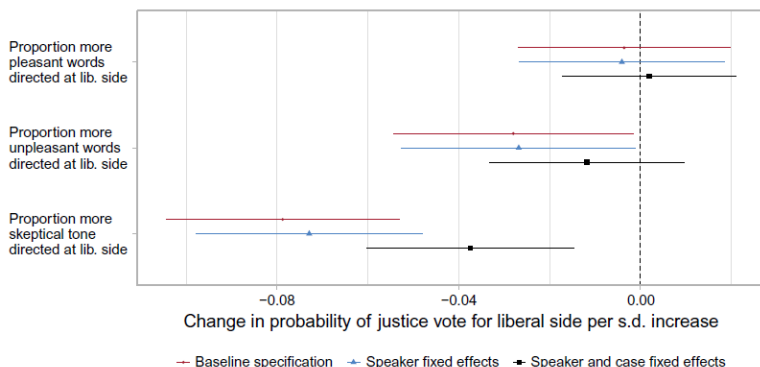
“Emotional Arousal” (Dietrich, Enos, & Sen 2019)

Table 1. Does vocal pitch predict votes in favor of the petitioner?

	Intercept only (1)	No controls (2)	DAL (3)	Harvard IV (4)	LIWC (5)
Fixed effects					
Constant	0.202 (0.056)	0.178 (0.055)	-0.025 (0.160)	-0.027 (0.160)	-0.026 (0.160)
Pitch difference		-0.266 (0.036)	-0.214 (0.038)	-0.215 (0.038)	-0.214 (0.038)
% More unpleasant words directed at petitioner			-1.971 (1.471)	0.071 (0.846)	-2.129 (1.308)
% More pleasant words directed at petitioner			-1.647 (1.086)	0.272 (0.685)	-1.673 (1.046)
# More questions directed at petitioner			-0.057 (0.008)	-0.057 (0.008)	-0.057 (0.008)
Political ideology _{t-1}			0.158 (0.033)	0.158 (0.033)	0.158 (0.033)
LC decision was conservative			0.011 (0.073)	0.012 (0.073)	0.012 (0.073)
Political ideology _{t-1} × LC decision was conservative			-0.263 (0.034)	-0.264 (0.034)	-0.263 (0.034)
Solicitor general as amicus supporting petitioner			0.540 (0.079)	0.543 (0.079)	0.540 (0.079)
Solicitor general as amicus supporting respondent			-0.672 (0.104)	-0.666 (0.104)	-0.673 (0.104)
# of amicus briefs supporting petitioner			0.039 (0.008)	0.039 (0.008)	0.039 (0.008)
# of amicus briefs supporting respondent			-0.058 (0.007)	-0.058 (0.007)	-0.058 (0.007)
Petitioner's level of resources			0.045 (0.014)	0.046 (0.014)	0.045 (0.014)
Respondent's level of resources			-0.003 (0.014)	-0.003 (0.014)	-0.003 (0.014)
Random effects					

“Model of Speech” (Knox & Lucas 2021)

FIGURE 4. Predicting Justice Votes with Directed Skepticism and Directed Affective Language



Note: Horizontal error bars represent point estimates and 95% confidence intervals from regressions of justice votes on directed pleasant words, directed unpleasant words, and our audio-based directed skepticism. Red circles correspond to a specification with no additional controls; blue triangles report results with speaker fixed effects only, black squares with speaker and case fixed effects.

Overview

1 Audio

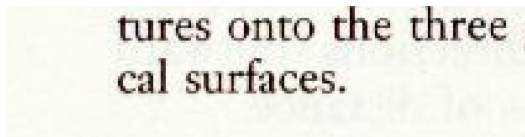
2 Image

3 Video

Images as Data

Images tend to be one of **two** types:

1. Raster images (pixels) – .bmp, .jpg, .jpeg, .png
2. Vector images (points, lines, and polygons) – .svg, .pdf, .eps



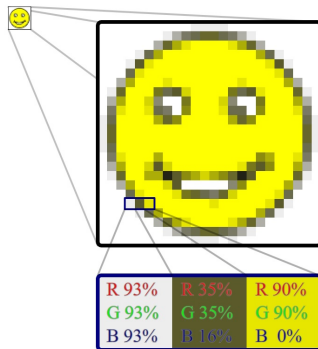
Source: <https://facul>

Note: Spatial data are actually a subset of image data!

Images as Data

We will focus on raster images, made up of three components:

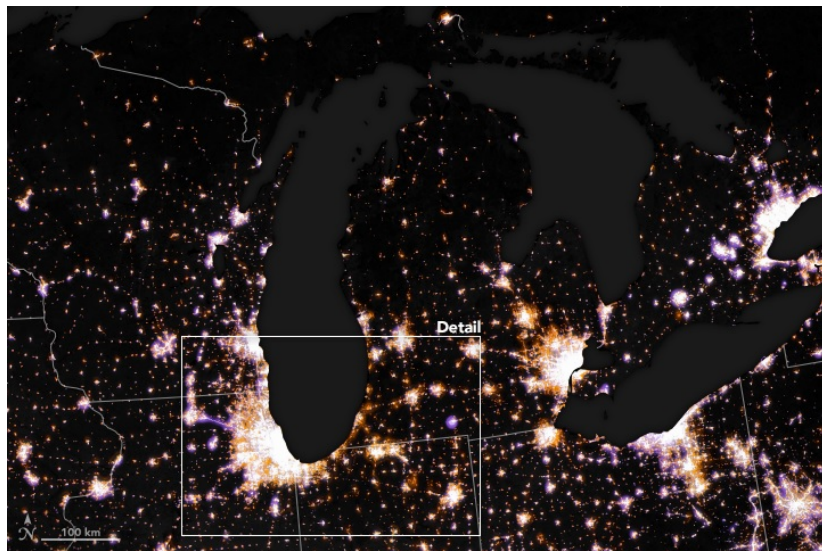
1. A grid of cells (equally sized)
2. A numerical value for each cell
3. A mapping of numerical values to colours (usually)



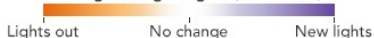
1. Remote sensing:

- **Electricity** (e.g., Baskaran, Min & Uppal 2014, Min et al 2013, Min et al 2014)
- **Deforestation** (e.g., Baragwanath & Bayi 2020, DiPoppa & Gulzar 2024)

Remote Sensing Data Example



Change in Night Lights (2012–2016)



Remote Sensing Data



Change in Night Lights (2012–2016)

Lights out

No change

New lights

“Local Inequality” (Weidmann & Theunissen 2021)

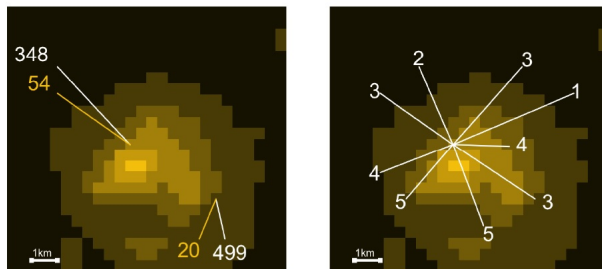


Figure 1. Satellite imagery of nighttime light emissions in raster format for the town Kansanshi in Zambia. **(Left panel):** The computation of the local Gini coefficient requires log-transformed nighttime light emissions at the level of cells (in yellow) and population estimates (in white). **(Right panel):** The DHS Wealth Index values of the households in the survey cluster at that location. All values are hypothetical and only displayed for illustration purposes.

“Deforestation” (Baragwanath & Bayi 2020)

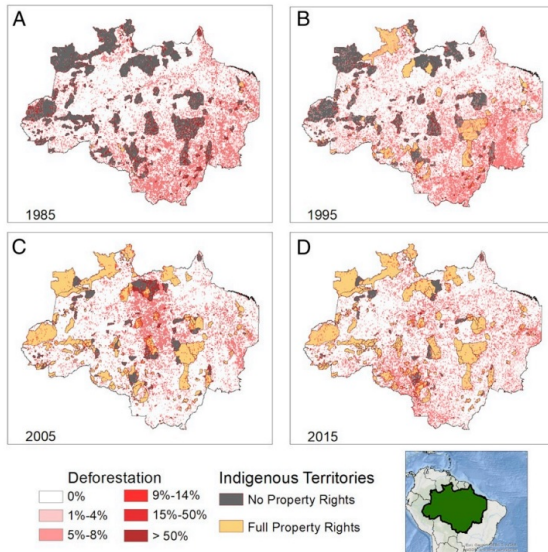


Fig. 1. Indigenous territories and deforestation in Brazilian Amazon for (A) 1985, (B) 1995, (C) 2005, and (D) 2015. Gray plots represent indigenous territories which have not received full property rights, while yellow plots represent plots which have received full property rights.

1. Remote sensing:

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2. Image classification tasks:

- **Classifying media imagery** (e.g., Torres 2024)
- **Detecting anomalies in photographs** (e.g., Cantú 2019)

“Bag of Visual Words” (Torres 2024)

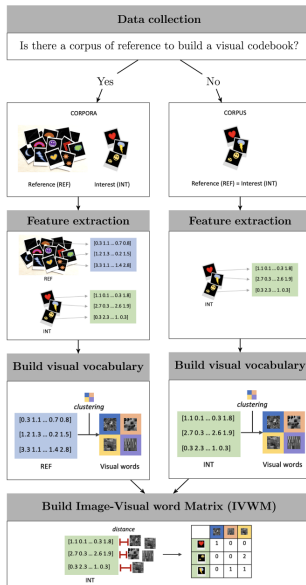


Figure 1. Workflow for building an Image-Visual Word Matrix.

“Bag of Visual Words” (Torres 2024)

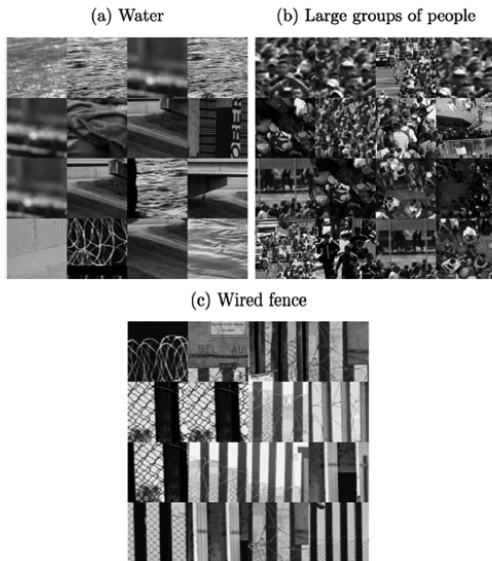


Figure 6. Examples of visual words.

“Fingerprints of Fraud” (Cantú 2019)

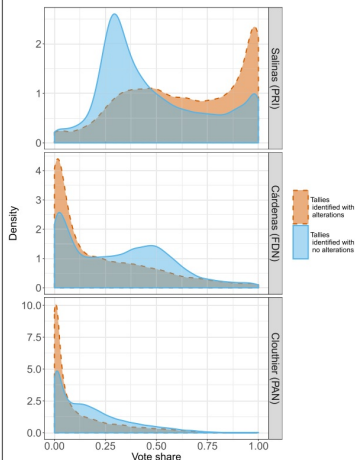
Francisco Cantú

FIGURE 1. Examples of Vote Tallies with Alteration in Their Numbers. Mexico, 1988



“Fingerprints of Fraud” (Cantú 2019)

FIGURE 4. Distribution of Vote Shares for Each of the Candidates. Mexico, 1988



Notes: The plots show the density distribution of the vote shares for the three main candidates of the 1988 election. Each line type corresponds to the classification of the vote tally sheet using the CNN classifier.

Overview

1 Audio

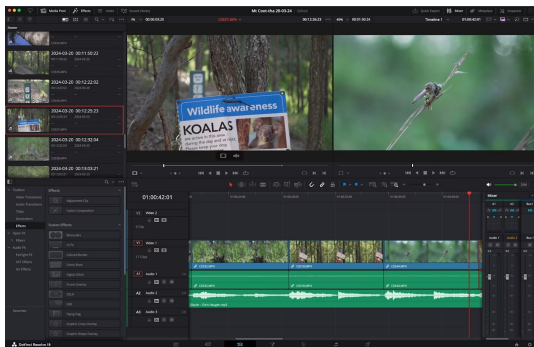
2 Image

3 Video

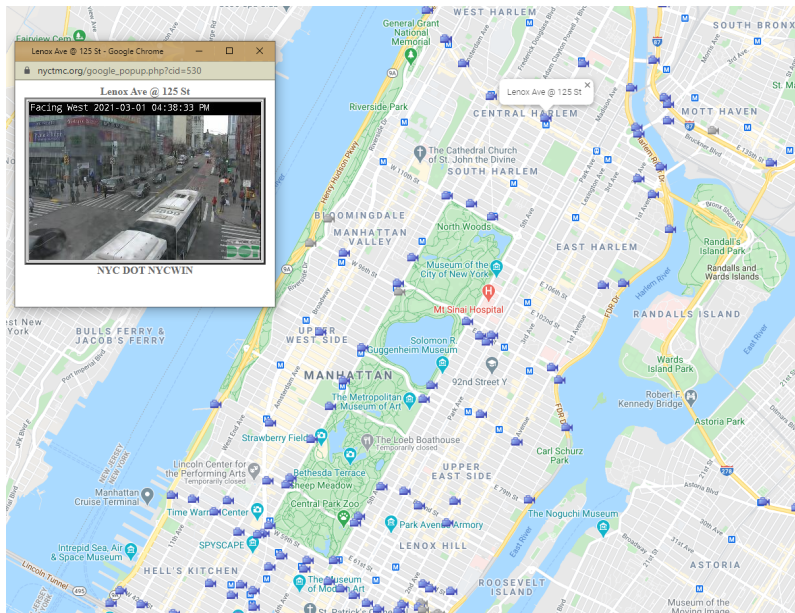
Video vs. Image/Audio Data

Video data represents a **combination of image** data and **audio** data:

- Sequence of still im-ages
- Sequence of sounds
- Played back at some known speed, in unison



“Racial Avoidance” (Dietrich & Sands 2023)



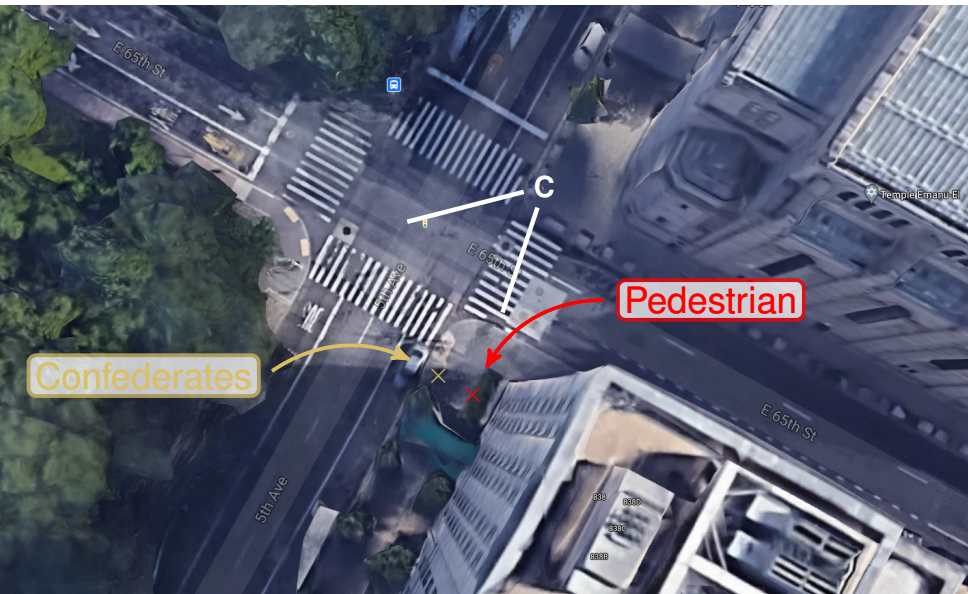
Facing East

2017-07-10 13:46:19



Pedestrian

Confederates



Facing East

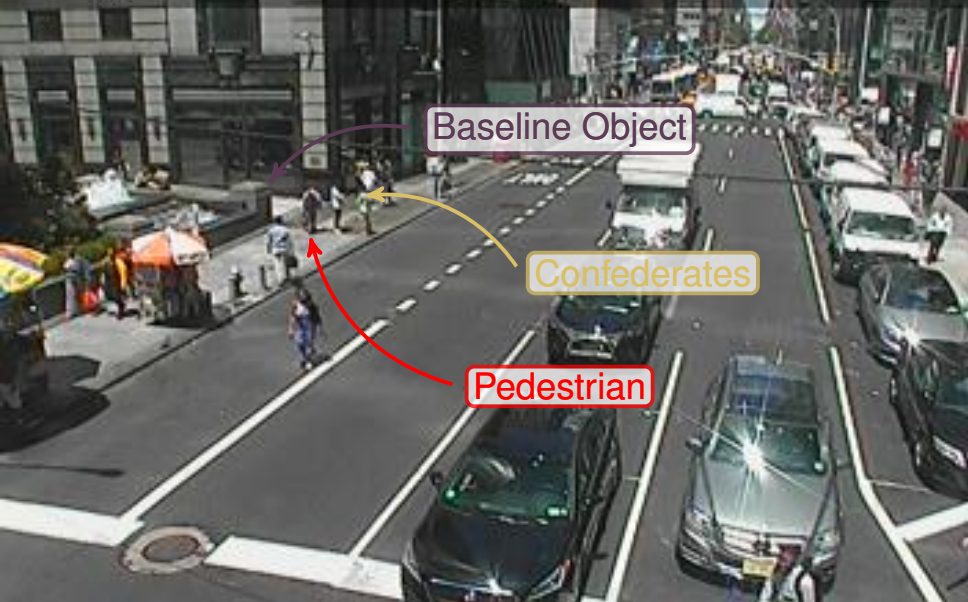
2017-07-10 13:46:19

Baseline Object

Pedestrian

Confederates

Facing East 06/20/2017 01:46:21 PM



From Video to Data

RAs record **pixel coordinates** for **each pedestrian**, **confederates**, and

imageJ Tracking

experiment	pedestrian_id	track	slice	x	y	block	condition	wall_distance	confederate_distance	distance_diff	inside_box
lexington	lexington_1	1	1	83	99	1	1	21.023796	38.013156	16.9893596	0

Facing East 2017-07-10 16:29:04

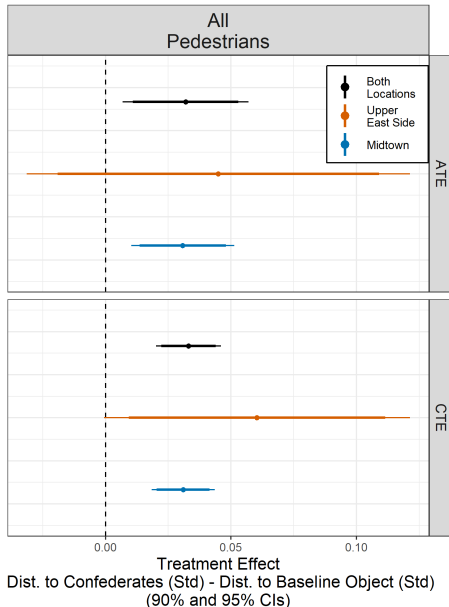


Facing East 06/20/2017 01:11:26 PM

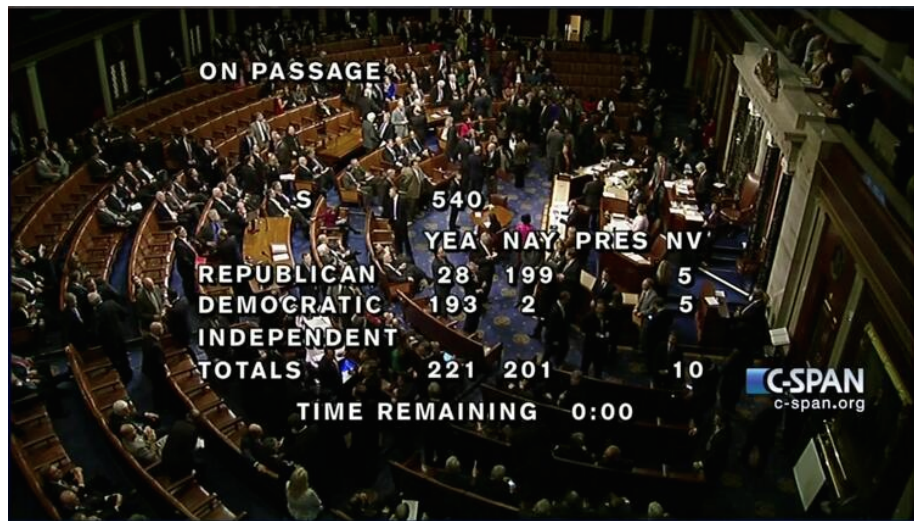


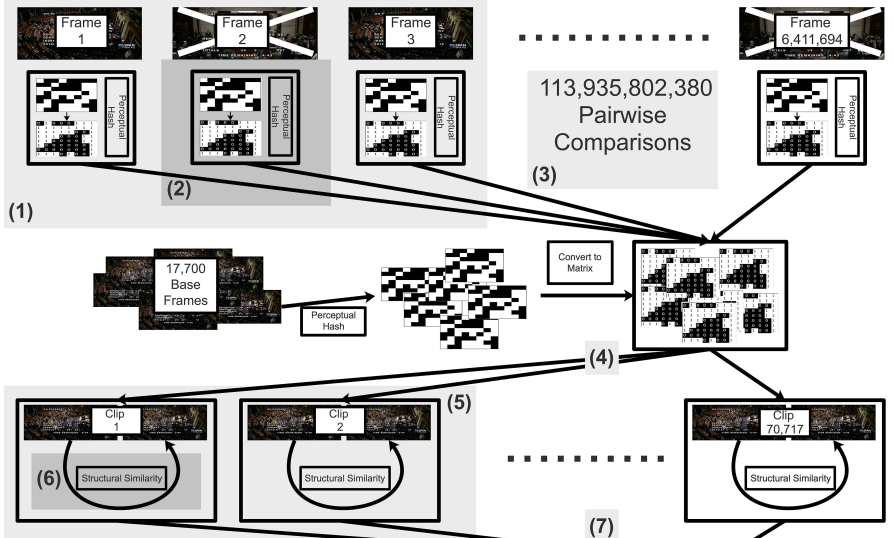
lexington	lexington_19	19	5	111	89	1	1	13.038405	14.866069	1.8276639	1
lexington	lexington_19	19	6	106	92	1	1	8.246211	17.000000	8.7537887	1

“Racial Avoidance” (Dietrich & Sands 2023)



"Crossing the Aisle" (Dietrich 2021)

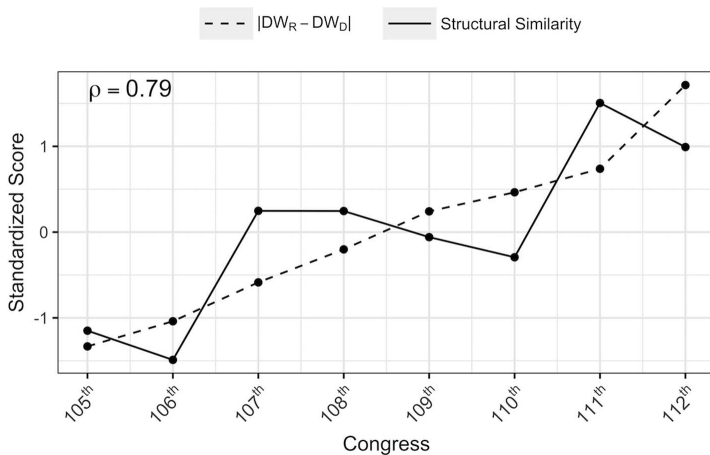




- (1) Using ffmpeg extracted frames from 6,526 C-SPAN videos (2009-2014).
- (2) Used a perceptual hashing algorithm to create a fingerprint for each frame.
- (3) Compared the fingerprints for all frames with fingerprints of 17,170 base frames.
- (4) If a frame was sufficiently similar to the base frames, then it was set aside.
- (5) Using sequential frames, created 70,710 clips and added meta data to each.
- (6) Used frame differencing and the Structural Similarity (SSIM) to calculate the aggregate motion for each clip.
- (7) Paired the clips with the corresponding roll call votes to create final dataset.

1	video	frame1	frame2	sim
2	house-vote-105-1-318	house-vote-105-1-318_f158.jpg	house-vote-105-1-318_f159.jpg	0.96414914
3	house-vote-105-1-318	house-vote-105-1-318_f162.jpg	house-vote-105-1-318_f163.jpg	0.96137325
4	house-vote-105-1-318	house-vote-105-1-318_f170.jpg	house-vote-105-1-318_f171.jpg	0.95554648
5	house-vote-105-1-318	house-vote-105-1-318_f171.jpg	house-vote-105-1-318_f172.jpg	0.96414914
6	house-vote-105-1-318	house-vote-105-1-318_f172.jpg	house-vote-105-1-318_f173.jpg	0.96414914
7	house-vote-105-1-318	house-vote-105-1-318_f174.jpg	house-vote-105-1-318_f175.jpg	0.96414914
8	house-vote-105-1-318	house-vote-105-1-318_f175.jpg	house-vote-105-1-318_f176.jpg	0.96414914
9	house-vote-105-1-318	house-vote-105-1-318_f176.jpg	house-vote-105-1-318_f177.jpg	0.96497345
10	house-vote-105-1-318	house-vote-105-1-318_f177.jpg	house-vote-105-1-318_f178.jpg	0.96497345

Polarization & SSIM Over Time (Dietrich 2020)



Dietrich, Bryce J. "Using motion detection to measure social polarization in the US House of Representatives." *Political Analysis* 29, no. 2 (2021): 250-259.

"LocalView" (Barari & Simko,)

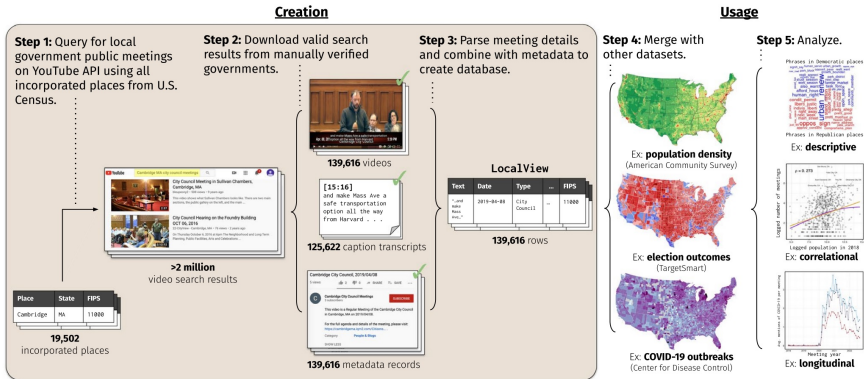


Fig. 2 Graphical Overview of Database Creation and Usage. Creation steps are performed on a regular cadence by the maintainers of the database; usage steps demonstrate how users may conduct analyses on the data, once it is downloaded.